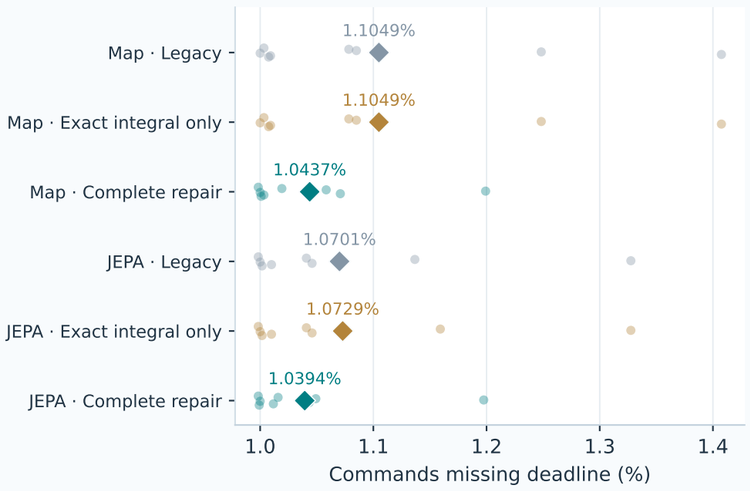
Fixed-weight JEPA: prospective controller and recovery repair

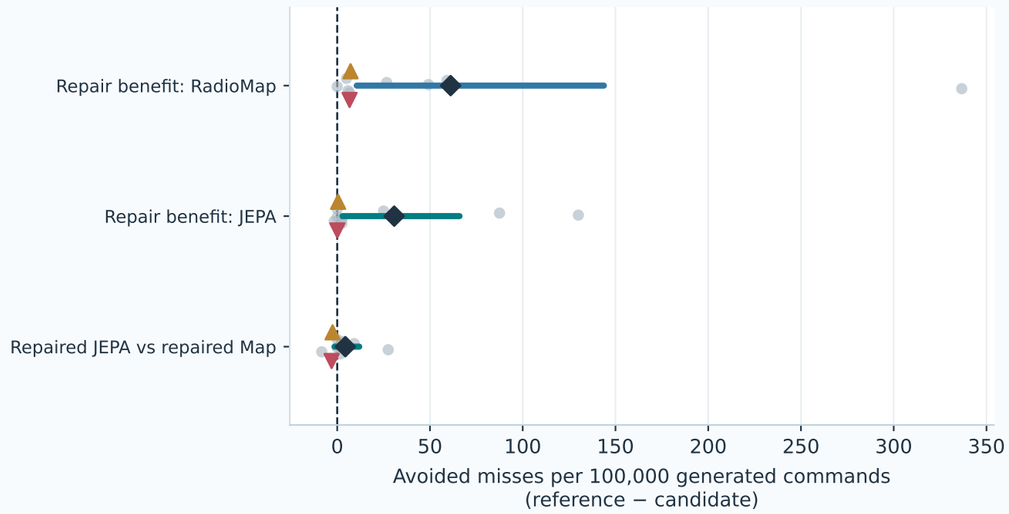
100 simulated robots × 60 s · 8 tested confirmation sites · 48 factual runs · no new fitting

Legacy and repaired controllers pay the same actual-cost forecast rule. Every registered comparison remains visible.

A Observed command loss
0.03 s handover assumption



B Paired service reduction
Positive favors the named candidate



C Registered gates

Statistics	Engineering	Benefit
PASS	FAIL	FAIL
PASS	PASS	PASS
FAIL	PASS	FAIL

D Development veto
4 s handover assumption

SEVERE-REGRESSION VETO

Two development sites; repair − legacy loss

RadioMap +5.505 pp · JEPA +4.748 pp

Registered stop threshold: +1.000 pp

4 s confirmation cases were not run.

E No confirmation estimates
Six-comparison budget retained

All three slow-cost primary contrasts remain untested

F Registered gates

Statistics	Engineering	Benefit
UNRUN	UNRUN	UNRUN
UNRUN	UNRUN	UNRUN
UNRUN	UNRUN	UNRUN

Paired whole-site observations · Mean reduction · Descriptive 95% site-bootstrap interval · Multiplicity-adjusted lower bound · Conservative endpoint lower bound

All six one-sided comparisons retain the 1 − 0.05/6 bound; both reduction lower bounds must exceed zero. Intervals use 20,000 whole-site bootstrap draws, conditional on fixed weights.
Pending commands are misses for the candidate and successes for the reference. Engineering tests cover worst-decile loss, silence, original video and bulk.
Source/causal/paid completion/offline replay integrity: True. Profile exceedances are paid capacity diagnostics under this NEW contract; older study vetoes remain unchanged.
Fast exact-integral-only arms are secondary ablations. Whole controller-package effects do not establish JEPA-objective necessity, hardware safety or SOTA.